

- 36** Gas exchange between air in the alveoli and blood in the capillaries is dependent on the process of diffusion.

The rate of diffusion is proportional to $\frac{\text{surface area} \times \text{concentration difference}}{\text{distance over which diffusion occurs}}$.

What would result in the fastest rate of diffusion?

- A** The distance is doubled, and the surface area is doubled.
- B** The concentration difference is halved, and the surface area is doubled.
- C** The distance is halved, and the surface area is doubled.
- D** The concentration difference is trebled, and the other variables are kept the same.